Assignment 1

Objective

To understand and practice data visualization concepts using the **Matplotlib** and **Seaborn** libraries by plotting mathematical functions, time-series data, multi-line graphs, shaded regions, dual-axis plots, and basic graphical shapes.

Dataset Description:

The Students Performance in Exams dataset contains exam scores of students in math, reading, and writing**,** along with demographic and background information such as gender, parental education level, lunch type, and test preparation course.

Dataset Link: *Students Performance Dataset*

Instructions:

1. Download the dataset from the link provided above.
2. Use Python's matplotlib library to create the following types of plots. Ensure that each plot adheres to the specified requirements.
3. Save each generated plot as an image file.
4. Compile all saved images into a single document and submit it.

Tasks:

1. Box Plot

   o Create a box plot to compare the distribution of math scores across different gender categories.

   o Show median, quartiles, and outliers clearly.

   o Title: "Math Score Distribution by Gender".

   o Label the x-axis as "Gender" and the y-axis as "Math Score".

2. Violin Plot

   o Create a violin plot to visualize the distribution of writing scores for different parental level of education.

   o Use different colors for each education level.

   o Title: "Writing Score Distribution by Parental Education".

   o Label the x-axis as "Parental Level of Education" and the y-axis as "Writing Score".

3. Heatmap

o Create a heatmap to display the correlation matrix of math, reading, and writing scores.

o Use a colorbar to represent correlation strength.

o Annotate each cell with correlation values.

o Title: "Correlation Heatmap of Student Scores".

4. Subplots (Dashboard View)

o Create a 2×2 subplot layout containing:

o Histogram of math scores.

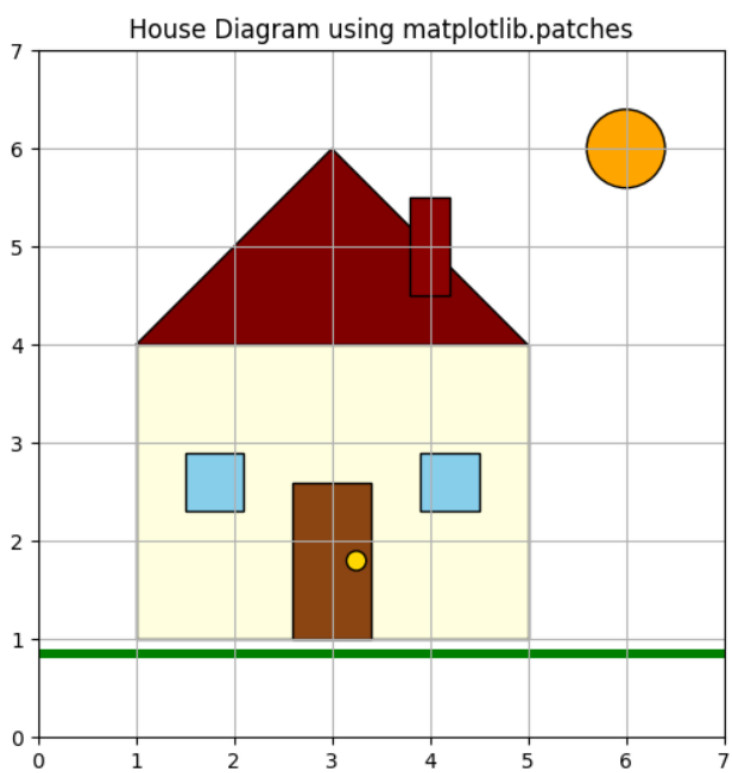
o Box plot of reading scores by gender.

o Scatter plot of math vs writing scores.

o Bar chart of average writing score by gender.

o Add a common figure title.

5. Draw a house diagram using matplotlib.patches



6. Generate a 3D surface plot for the mathematical function:

$$z = x^2 + y^2$$

Use proper axis labels and title.

7. Using the matplotlib.image sub-library, perform the following operations on a single input image and display all outputs appropriately:

Load and Display Image

a. Read the image using matplotlib.image.imread().
b. Display the original image.

Convert to Grayscale

c. Convert the loaded image into grayscale format.
d. Display both original and grayscale images using subplots.

Image Manipulation

e. Crop a selected region of the image.
f. Flip the image horizontally.
g. Display original, cropped, and flipped images in a single figure.

Brightness and Contrast Adjustment

h. Increase the brightness of the image.
i. Modify the contrast of the image.
j. Display original and modified images side by side for comparison.

Watermarking

k. Overlay a transparent watermark text on the original image using Matplotlib functions.
l. The watermark should appear semi-transparent and positioned at a visible location.

Submission Requirements:

1. Save all the plots generated as image files (e.g., PNG, JPG).
2. Compile the images into a single Word or PDF document, including titles and brief descriptions for each graph.
3. Submit the document as per the instructor's instructions.
4. Organize exercises clearly under headers.
5. Include question, code, output, and short explanations as comments.
6. Header on each page must be

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7. Footer on each page must be

Additional Notes:

- Ensure proper labeling of axes and titles for all plots.
- Use appropriate colors and styles for better visual clarity.
- Verify that all graphs are legible and accurately represent the data.